Dynamics of Premixed H_2/CH_4 Flames Under Near Blowoff Conditions

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Swirling flows are widely used in industrial burners and gas turbine combustors for flame stabilization. Several prior studies have shown that these flames exhibit complex dynamics under near blowoff conditions, associated with local flamelet extinction and alteration in the vortex breakdown flow structure. These extinction events are apparently due to the local strain rate irregularly oscillating above and below the extinction strain rate values near the attachment point. In this work, global temporally resolved and detailed spatial measurements were obtained of hydrogen/methane flames. Supporting calculations of extinction strain rates were also performed using detailed kinetics. It is shown that flames become unsteady (or local extinctions happen) at a nearly constant extinction strain rate for different hydrogen/methane mixtures. Based upon analysis of these results, it is suggested that classic Damköhler number correlations of blowoff are, in fact, correlations for the onset of local extinction events, not blowoff itself. Corresponding Mie scattering imaging of near blowoff flames also was used to characterize the spatio-temporal dynamics of holes along the flame that are associated with local extinction.
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1 Introduction

Swirling flows are widely used in industrial burners and gas turbine combustors for flame stabilization. A number of operational and performance issues are of concern in modern highly tuned dry low NO_x (DLN) combustion systems utilizing swirl stabilized flames, including blowout, flashback, combustion instabilities, and emissions [1,2]. The issue of lean blowoff is the focus of this study, a concern due to the requirement to operate DLN systems under very fuel lean conditions and, therefore, close to their blowoff limits. Flames can only be stabilized in high velocity reactant streams over a certain range of conditions. Determining these conditions, and those where the flame cannot be stabilized—referred to here as blowoff—is an important issue in any practical combustion device.

A large literature on predicting, measuring, and correlating blowoff limits already exists [3–6]. Several different theories or physical considerations have been used in past blowout correlation studies, such as those of Zukoski and Marble [7], Spalding [8], Longwell [9,10], and others. As noted by Glassman [4], however, they lead to essentially the same form of correlation that relates the blowoff limits to a Damköhler number, i.e., ratio of a fluid mechanic and chemical kinetic time, τ_{res}/τ_{chem}. The work of Hoffman et al. [3] is of particular interest here, as it demonstrated that related scaling could be used to correlate blowoff limits in swirling premixed flames.

We next briefly review several prior phenomenological characterizations of blowoff. Longwell et al. [9] suggested that blowoff occurs when it is not possible to balance the rate of entrainment of reactants into the recirculation zone, viewed as a well-stirred reactor, and the rate of burning of these gases. Since entrainment rates scale as D/U [11], this criterion reduces to a Damköhler number blowoff criterion, using a chemical time that is derived from the well-stirred reactor, τ_{PSR}. A similar idea relates to an energy balance between heat supplied by the hot recirculating flow to the fresh gases and that released by reaction [12–15]. In this view, blowoff occurs when the heat required by the combustible stream exceeds that received from the recirculation zone. This leads to the same entrainment based, fluid mechanical time scaling as above, and the resultant similar Damköhler number blowoff criterion.

A different view is that the contact time between the combustible mixture and hot gases in the shear layer must exceed a chemical ignition time [8,15–17]. For example, Zukoski [16,18] suggested that ignition of the incoming fresh unburnt mixture occurs in the shear layer as it mixes with combustion products from the recirculation zone behind the bluff body. The relevant chemical time in this description is an ignition time. The fluid mechanic time scaling associates the characteristic dimension by the recirculation zone length, which also being proportional to the bluff body size, leads to a similar Damköhler number correlation.

Finally, several studies have proposed a flamelet based description based on local extinction by excessive flame stretch [19,20].

It is difficult to determine which of the above descriptions most accurately describes the controlling processes based on analyses of *Da* correlations alone because the different velocity, length, and chemical time scales generally lead to comparable groupings of the data. However, improvements in fundamentals of turbulent combustion provide further insight for evaluation of these different blowoff descriptions. As emphasized by Driscoll [21] in a recent review, there is little evidence of the existence of distributed combustion or well-stirred reactors. Rather, modern diagnostics and computations have demonstrated that flamelets exist under a very broad range of conditions; reaction regions deemed to be spatially distributed from line of sight images have often been shown to consist of highly contorted three-dimensionally oriented reaction sheets when visualized with planar laser imaging techniques. This point is illustrated in Fig. 1, which illustrates two OH planar laser induced fluorescence (PLIF) images and a single short time exposure line of sight image of the same swirling flame. There is some possibility of the existence of a "thickened flamelet" regime, where turbulent diffusivity plays a role due to "stirring" by Kolmogorov scale eddies in the preheat zone, but evi-

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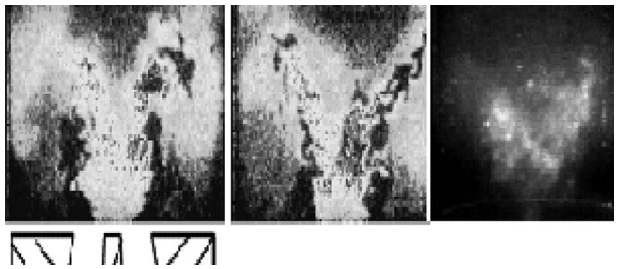

Fig. 1 Two instantaneous OH PLIF images of acoustically forced swirling flame (left and middle) and associated line of sight image (right). Flow bottom to top. Reproduced from Ref. [22].

dence is lacking at this point [21]. As such, blowoff theories postulating the existence of well-stirred reactor regimes, such as in the recirculation zone, probably do not correctly capture the controlling processes. Furthermore, a critical component to several of the above described blowoff theories is heat exchange from the hot wake to the reactants. However, given the fact that a premixed flame acts as an interface between reactants and products, there is no "contact" between the wake zone and fresh reactants, except for the thermodiffusive processes (and possibly mixing by Kolmogorov scale eddies) confined to the preheat zone of the flamelet. In other words, no such exchange occurs outside of points very near the attachment region, except for flames near blowoff where extensive holes in the flame sheet exist.

Given these points, the subsequent discussion will focus on flamelet based descriptions of the controlling blowoff processes, which are also captured by Damköhler number scalings. Extinction of flamelets occurs through two mechanisms: heat transfer and flame straining [23,24]. Flame strain is apparently the dominant effect, with heat/radical losses only dominant very near the bluff body. Flame strain occurs due to gradients in flow velocity, such as if the flame resides in a shear layer, or flame curvature, such as when it is rolled up in a vortex. Flames are only capable of withstanding certain levels of strain before extinction, denoted as the extinction strain rate, κ_{ext}. As detailed in Ref. [23], the extinction strain rate is a function of fuel composition (particularly its molecular weight relative to air), stoichiometry, and chemical time scale. The inverse of the extinction strain rate forms a chemical time, $\kappa_{\text{ext}}=1/\tau_{\text{ESR}}$, whose relationship to other kinetic time scales is discussed in a number of references [25–27].

The local Damköhler number definition used in this paper can be written as a ratio of the extinction flame strain rate and the flow strain rate

$$Da=\frac{\tau_{\text{flow}}}{\tau_{\text{ESR}}}=\frac{\kappa_{\text{ext}}}{\kappa} \tag{1}$$

where κ denotes the reference strain rate, inversely proportional to the reference fluid mechanical time, τ_{flow}.

As suggested in several prior studies, *it is clear that flamelet extinction occurs near blowoff, but does not immediately lead to blowoff* [28,29]. In other words, blowoff does not occur the instant when $\kappa_{\text{ext}}=\kappa$ at some point along the flame. As such, the presence of strain induced extinction of the flame sheet near blowoff conditions, and the fact that the *Da* describes such extinction, is useful, but not a complete description of the blowoff phenomenon.

Local extinction is known to lead to substantial unsteadiness in flames near blowoff [20,30–33]. Holes on the flame sheet are initiated at points of high flame strain. This topic has received extensive discussion in the more general turbulent combustion literature [34–36]. The formation of "edges" that are associated with holes on the flame sheet is a more general combustion problem that has received recent attention for both premixed and nonpremixed flames [37–41], including a recent review by Buckmaster [42]. Once a flame hole is initiated by excessive local strain/scalar dissipation rates, it can stay the same size, grow, or shrink, associated with whether the edge flame stays stationary, retreats, or advances into the hole, respectively. In addition, the hole is convected with the mean flow at the local tangential mean flow velocity, as will be discussed further in this paper. For premixed flames, it is important to note that the advance/retreat velocity of the flame edge is not related to the laminar burning velocity. Furthermore, while flame holes appear at points where the local strain rate, κ, exceeds κ_{ext} (in the quasisteady case), the flame edge does not correspond to the point in the flow where $\kappa=\kappa_{\text{ext}}$. Rather, Liu and Ronney's results [41] show that the flow strain rate at the flame edge value, κ_{edge}, is lower than κ_{ext} by a factor of up to two (i.e., $0.5<\kappa_{\text{edge}}/\kappa_{\text{ext}}<1$). In other words, once a hole is initiated in a region of high strain, it can lead to flamelet extinction at adjoining points that would otherwise not have extinguished at the local conditions.

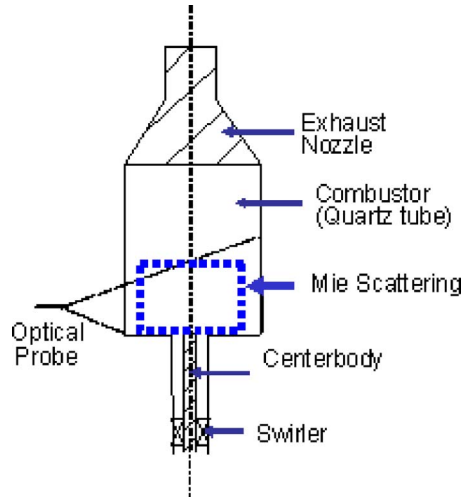

Fig. 2 Schematic of the combustor facility with Mie measurement window and optical probe

It has been suggested that flames approach blowoff in two phases, the first characterized by local extinction on the flame sheet, and the second by large-scale alteration of the fluid mechanics. The objective of this study is to characterize these dynamical blowoff processes in this first phase. This study closely follows several related studies of our group, which have characterized the dynamic blowoff process of pilot, bluff body, and swirl stabilized flames [29,43]. In the swirling flame, Muruganadam et al. [29] and Zhang et al. [44] showed that the swirling flame tends to oscillate between extinction and re-ignition phases. The number of extinction/re-ignition events per unit time monotonically grows as blowoff is approached. Very near blowoff, the entire structure of the vortex breakdown appears to change in a complex manner, with various helical flow features appearing and reappearing in a sporadic fashion.

2 Instrumentation, Experimental Facility, Experimental Procedures, and Data Reduction

Measurements were obtained in a lean premixed swirl combustor, which was duplicated from an experimental rig at Sandia National Laboratories [45]. This was done in order to have the same test facility to facilitate comparisons of data and simulations with the Sandia group. The combustor is schematically shown in Fig. 2. The facility consists of a swirler/nozzle, combustor, and exhaust sections. Premixed gas, consisting of H_2/CH_4 mixtures, and air flows through a swirler/nozzle section. The nozzle is an annular tube with inner diameter of 28 mm. The center body has an outer diameter of 20 mm. The overall flow area remains constant at 3.0 cm^2 inside the nozzle. Tests were performed with a six-vane

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45 deg swirler, which is located in the annulus between the centerbody and nozzle wall. The theoretical swirl number, which is 0.85, is calculated from the relation [46]

$$S = \frac{2}{3}\left[\frac{1-(d_h/d)^3}{1-(d_h/d)^2}\right]\tan\theta \tag{2}$$

where d_h and d are the diameters of centerbody and swirler, respectively, and θ is the swirler vane angle. The fuel is injected 150 cm upstream of the combustor to achieve a premixed condition. The combustor consists of a 305 mm (12 in.) long quartz tube, with a 115 mm and 120 mm inner and outer diameter, respectively. It rests in a circular groove in a base plate. An exhaust nozzle has a 152 mm contraction section with the area ratio 0.44, and a 102 mm long, 51 mm inner diameter chimney section.

The air and fuel flow rates are measured with a flowmeter and mass flow controllers (MFCs), respectively. Both the flowmeter and MFCs were calibrated using the specific gas with which they were to meter. The maximum resultant uncertainty in ratios is 0.01–0.02 for most of the cases. The largest uncertainty in ϕ of 0.03 occurs with pure CH_4. The air is choked before the mixing section, and the premixed air/fuel is choked again inside the inlet tube of the combustor (not shown) upstream of the swirler to minimize the impact of perturbations in the combustor impacting the fuel/air mixing process.

All experimental measurements were obtained at a constant nozzle exit velocity of 33 m/s. Tests were performed at a combustor pressure of 1.0 atm and 300 K reactants.

The Mie scattering test in the combustor was performed using particle image velocimetry (PIV) facility. The system consists of a dual head Nd:YAG laser, a high resolution charge coupled device (CCD) camera, a mechanical shutter, and a centralized timing generator. In addition a cyclone seeder was used to supply anhydrous aluminum oxide (Al_2O_3) with an average particle size of 3 μm.

Each laser head delivered a 5 mm 120 mJ/pulse beam at a wavelength of 532 nm. The beams pass though an optical arrangement for generation of a light sheet with desired height and thickness. This arrangement consisted of a convex spherical (f=1 m) and a convex cylindrical lens (f=25.4 mm) that resulted in a light sheet 1 mm thin at the centre of the combustor. The CCD camera captured the images of the illuminated particles at a resolution of 1600×1200 pixels (corresponding to 52×39 mm^2) in frame straddling mode. The duration between laser shots was set at 5 μs. In addition the camera was also fitted with a 532 nm laser line filter with a FWHM of 3 nm to restrict any background noise. For each image group (two images), the first image was used.

In addition, UV radiation from flames was monitored with an optical fiber bundle (NA=0.44), with the head located 46 mm above the dump plate of the combustor and 171 mm radially from the combustor centerline, see Fig. 2. This volume was placed such that light is collected from the lower one-third of the combustor, in order to image the inner recirculation zone (IRZ). The light passes through an interference filter centered at 308 nm and with a full-width-half-maximum (FWHM) of 10 nm, which corresponds to the primary spectral region of OH^* emission. This radiation was detected by a miniature metal package photomultiplier tubes (PMT) (Hamamatsu H5784-04). This PMT has a built-in amplifier (bandwidth of 20 kHz) to convert the current to voltage and operates from a 12 VDC source.

The signal output from the sensors was low pass filtered by a Krohn-Hite Model 3362 digital Butterworth filters and then fed into a National Instruments A/D board. The sampling frequency was 2 kHz. The low pass filter frequency (for anti-aliasing) was set at half the sampling frequency, 1 kHz.

3 Results and Discussion

3.1 Global Dynamics-Observation. The blowoff limits for this combustor are indicated by the dashed red line in Fig. 6.

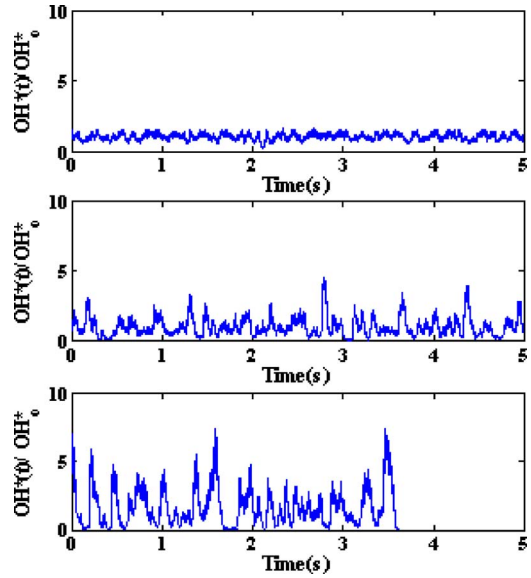

Fig. 3 Time series data of OH^* signal of CH_4 flame at equivalence ratios of 0.8 (top), 0.62 (middle), and 0.51 (bottom), where OH^*_o denotes time average of $OH^*(t)$

However, the flame becomes unsteady and exhibits transient behaviors before blowoff. This section presents typical results of these flames under near blowoff conditions.

We first present "global," but time resolved characterization of near blowoff dynamics. Figure 3 plots the OH chemiluminescence signal from the optical probe (see Fig. 2) at three equivalence ratios approaching the blowoff value. There is clearly a significant change in characteristics of these time series as blowoff is approached.

A detail of the time series for a near blowoff case is shown in Fig. 4. The regions associated with a drop and rise in chemiluminescence signal are referred to here as "events," which are interpreted as local extinction and/or convection of the flame downstream, followed by re-ignition and/or propagation of the flame into the mass of unburned reactants, which accumulate due to the flame moving downstream. Following the method of Muruganandam et al. [29], events are defined here as initiating at the point in time where the intensity of the signal drops lower than some threshold, and ending when the signal goes above a second higher

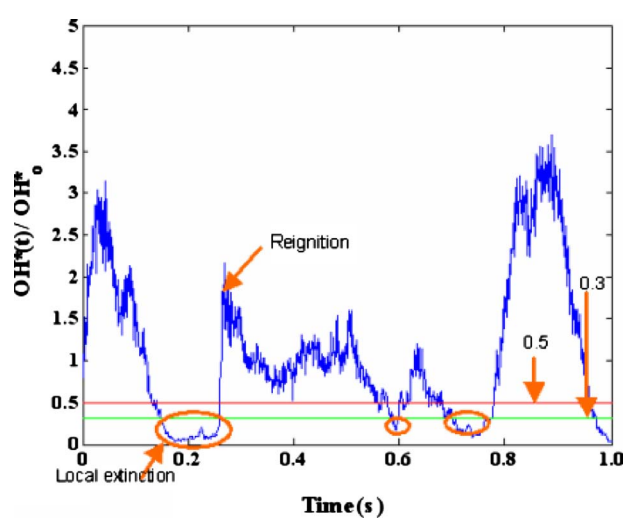

Fig. 4 Time series data of OH signal for methane flame at equivalence ratio of 0.54

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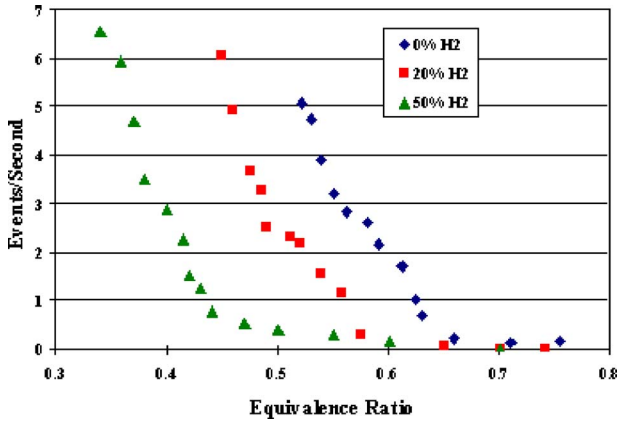

Fig. 5 Dependence of local extinction event frequency on equivalence ratio of CH_4/H_2 flames

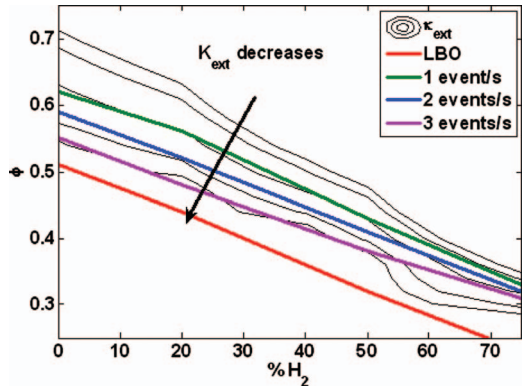

Fig. 6 Dependence of κ_{ext}, blowoff limits, and event rate on percentage of hydrogen and equivalence ratio. Contour lines of extinction strain rate are indicated at 1400 1/s, 1200 1/s, 800 1/s, 400 1/s, and 200 1/s. These contours were estimated by calculating κ_{ext} at 0/100%, 20/80%, 40/60%, 50/50%, and 75/25% H_2/CH_4 mixtures with equivalence ratio steps of 0.02. Blowoff limits and event information (dash lines) were collected for 0/100, 20/80, 50/50, and 75/25 H_2/CH_4 mixtures.

valued threshold. This second threshold value is needed to eliminate multiple counts of the same event that oscillates above and below the same threshold. For these data, 0.3 and 0.5 (of the mean) were used as the first and second thresholds for local extinction. These distinctive extinction and re-ignition events span a period from O(1s) to O(0.001s), without any obvious periodicity or frequency prior to blowoff. As the LBO limit is approached, more of these events occur in a given time period and thus the time between successive events decreases.

The average number and duration of events per unit time can be quantified for a given set of prescribed threshold values. Typical results are shown in Fig. 5 as a function of equivalence ratio for three CH_4/H_2 mixtures. The test was initiated under stable conditions with the equivalence ratio decremented over steps of 0.01, with 30 s of data taken at each point all the way to blowoff. These sweeps were repeated three times and the results averaged. The near zero value of the event count far from blowoff, the "knee in the curve" associated with the initiation of events, and the monotonic rise in event count with further reduction in fuel/air ratio are evident in the figure. Similar results are exhibited for three fuel blends, with the higher hydrogen ones shifted towards lower equivalence ratios, as expected [6], but the curves have similar characteristics. The absolute value of equivalence ratio at which events begin is a function of the specific threshold values used to flag events, but does not change the trends shown in this figure.

An alternative way of plotting these data is indicated by the dashed lines in Fig. 6. This figure plots lines of constant event count as a function of equivalence ratio and percentage of H_2 in the fuel. These data illustrate that these lines are roughly parallel to each other.

3.2 Global Dynamics-Discussion. This section describes an analysis of the above data using computed chemical times. Three chemical times were calculated: (1) the "blowoff residence time" of a perfectly stirred reactor, τ_{PSR}, (i.e., the minimum residence time for which non-negligible reaction progress occurs) (2) unstretched laminar premixed flame time scale, given by the ratio of the premixed flame thickness and flame speed, α/S_L^2. (3) inverse of the extinction strain rate of an opposed flow, laminar premixed flame, $\tau_{ESR}=1/\kappa_{ext}$, and the first two times were determined using CHEMKIN software tools PSR and PREMIX, respectively. Extinction strain rates were calculated using an arc-length continuation method, implemented in COSILAB 2.0 using the GRI 3.0 mechanism.

Referring back to Fig. 6, computations of extinction strain rate were also performed for similar ranges of ϕ and CH_4/H_2 ratios. These calculations were interpolated and used to plot isolines of extinction strain rates, indicated by the solid black lines. Since all geometric and dimensions and flow velocities were fixed, these lines also correspond to lines of constant global Damköhler number, as will be discussed in the context of Fig. 7.

Significantly, these data show that these iso-event count lines are parallel to constant Damköhler number lines, showing that each Damköhler number can be associated with a particular event rate. Define the flow time in the Damköhler number definition, see Eq. (1), as D/U_o, where D indicates the diameter of the centerbody and U_0 is the flow speed at nozzle exit. With this definition, event counts of 1 events/s, 2 events/s, and 3 events/s correspond to $Da=0.5$, 0.35, and 0.2, respectively.

Using these ideas, the data shown in Fig. 5 can be replotted by replacing the x-axis with the calculated Damköhler number, see Fig. 7. This figure shows a very similar relationship between Damköhler number and event rate across the three different fuel blends. Note that the lowest equivalence ratio data shown in Fig. 5 is not presented in Fig. 7 due to difficulties in obtaining converged chemical time solutions at these points.

These data can also be used to support the argument that the extinction strain rate, $\kappa_{ext}=1/\tau_{ESR}$, as opposed to other kinetic time scales, is the most physically meaningful in describing these near blowoff dynamics. In many cases, different kinetic time

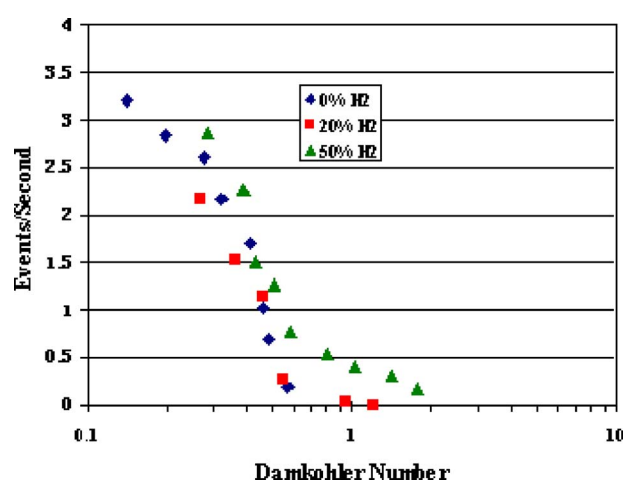

Fig. 7 Dependence of extinction event rate upon Damköhler number of CH_4/H_2 flames

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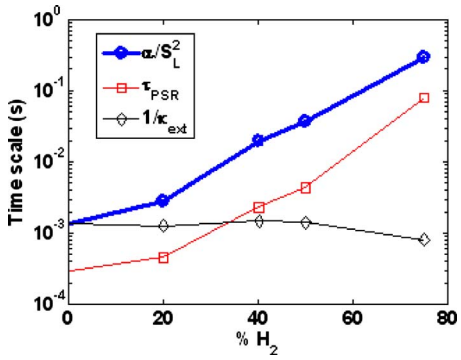

Fig. 8 Comparison of three chemical time scales at fuel/air ratios associated with event rate of 1 event/s

scales, such as those based on τ_{PSR} or ignition times, are closely related and, thus, give comparable Damköhler scaling. However, there are substantial variations in chemical time relationships for CH_4/H_2 mixtures with H_2 levels greater than about 20%, particularly between extinction strain rate based time scales and unstrained flame or PSR based time scales. Note that the latter is purely a kinetic time scale, while the former two time scales also depend on diffusive processes. Given the substantial difference in diffusivity of the fuel relative to oxidizer with increasing H_2 levels, it then follows that appropriate choice of time scale becomes increasingly important with high H_2 fuels. To restate, systematic differences between different kinetic time scales can be anticipated when comparing over a range of fuels with different diffusivities. This point is born out in this data. Figure 8 compares three calculated chemical times for near blowout conditions at constant event rates for CH_4/H_2 flames. Again, nozzle velocity and geometry were fixed so it is reasonable to presume relatively constant fluid mechanic time scale. As such, the y-axis on this graph is inversely proportional to the global Damköhler number. These data show that the extinction based kinetic scale is roughly constant across the whole range of H_2 levels, while the PSR and flame propagation based scales are not.

Further insight into these results can be obtained by distinguishing between a global average Damköhler number, $\overline{Da}$, such as used in plots in the Results section, and the actual instantaneous Damköhler number at the flame.[1] The actual strain rate is a fluctuating quantity with a mean and variance given by $\bar{\kappa}$ and σ_k. For a given $\overline{Da}$ and strain fluctuation amplitude and time scale, the flame sheet at a given point will be extinguished for some fraction of time, ζ. This is illustrated in Figure 9, which reproduces a plot from Shanbhogue et al. [28]. This figure plots a notional time series of the instantaneous Da value at some position, as well as the extinction value and time averaged value, $\overline{Da}$. The vertical highlighted stripes correspond to regions where the instantaneous Da value falls below the extinction value. If $\overline{Da}$ is decreased (e.g., by decreasing fuel/air ratio), the flame sheet is extinguished at this point for a larger fraction of time, illustrated by comparing the top and bottom figures. The fraction of time, which the flame is extinguished, will be denoted as ζ

Consider this extinction fraction, ζ, in further detail. Figure 10 overlays a hypothetical probability density function of the flame strain rate, with the extinction strain rate at a particular point along the flame, s. For illustrative purposes, it is assumed that κ_{ext} is a single number, not a distribution itself and that extinction occurs at points where the local strain rate, κ, exceeds the extinction strain rate, κ_{ext}. The latter assumption assumes a quasilocal[2] and quasisteady[3] flamelet [47]. The fraction of time that the flame is extinguished is given by the hatched area under the curve. This is given by the integral

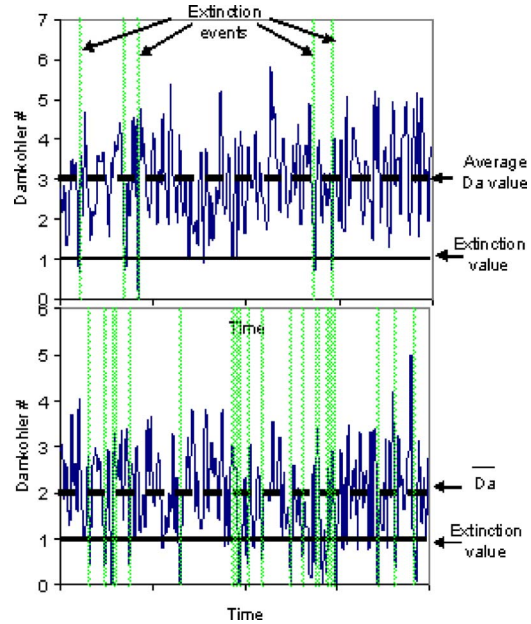

Fig. 9 Notional description of time variation of local Damköhler number at given point on flame sheet, illustrating local extinction events at two average Damköhler number values, $\overline{Da}$, one farther (top) and closer (bottom) to blowoff

$$\zeta(s) = \int_{-\infty}^{\kappa_{ext-}} P(\kappa,s)d\kappa + \int_{\kappa_{ext+}}^{\infty} P(\kappa,s)d\kappa \tag{3}$$

where κ_{ext-} and κ_{ext+} denote the negative and positive extinction strain rates. In order to see the form of the results, consider a Gaussian probability density function (PDF) (in general, not a good representation of strain rate PDFs, but one that allows for illustrative analytical treatment)

$$P(\kappa) = \frac{\exp[(\kappa-\bar{\kappa})^2/2\sigma_k^2]}{\sqrt{2\pi\sigma_k^2}} \tag{4}$$

Assuming equality of κ_{ext-} and κ_{ext+}, setting $\bar{\kappa}=\kappa_{ref}$, integrating, and rearranging the resulting expression to solve for the average Damköhler number, $\overline{Da}=\kappa_{ext}/\bar{\kappa}$, leads to

$$\overline{Da}(s) = \kappa_{ext}/\bar{\kappa} = 1 + \sqrt{2}\frac{\sigma_\kappa(s)}{\kappa(s)}\mathrm{erf}^{-1}[1-\zeta(s)] \tag{5}$$

Or equivalently, the local extinction fraction can be expressed in terms of $\overline{Da}$ and $\sigma_\kappa/\bar{\kappa}$ as

$$\zeta(s) = 1 - \mathrm{erf}\frac{\overline{Da}(s)-1}{\sqrt{2}\sigma_\kappa(s)/\bar{\kappa}(s)} \tag{6}$$

These expressions relate the fraction of the flame sheet that is locally quenched for a given average Damköhler number and strain rate fluctuation. They nicely mirror the experimental data shown earlier which show that each average Damköhler number is

[1]Much of the discussion in this section closely follows arguments from Ref. [28], made in the context of bluff body flames.

[2]This quasilocal treatment of the flame holes neglects the fact that once a flame hole is initiated, it can advance into a region with a strain rate that is lower than κ_{ext}, as discussed earlier.

[3]This reasoning can also be generalized to the non quasisteady flamelet case. A more general parameterization of flame extinction and strain rate is with functions of the form $\kappa_{ext}=f(\kappa,\dot{\kappa},\ddot{\kappa},\ldots)$ and $P(\kappa,\dot{\kappa},\ddot{\kappa},\ldots)$. We will not carry through the details of the calculation, but a similar integration can be performed.

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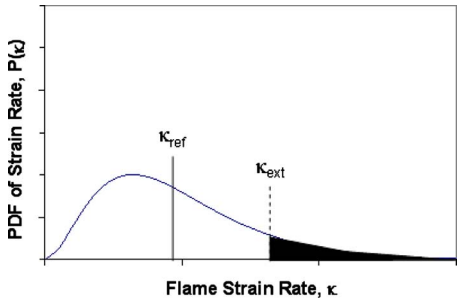

Fig. 10 Notional PDF of flame strain rate and extinction strain rate at a particular point along the flame, *s*

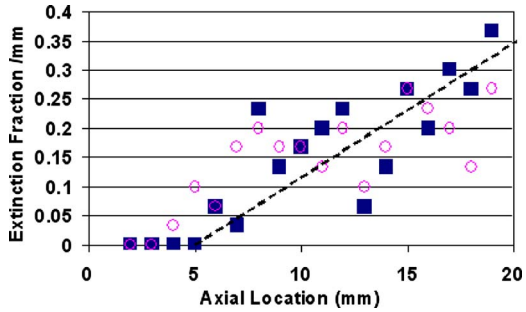

Fig. 12 Distribution of holes along the flame front for CH_4 (■) and 50/50 $H_2/CH_4(O)$ flame at equivalence ratio of 0.62 and 0.43, respectively. Line denotes calculated result assuming holes are generated uniformly and randomly along the flame and propagate downstream with the speed of 30 m/s.

associated with a given event count rate. Our bluff body blowoff review suggested that from a fundamental point of view, the average Damköhler number should not be considered as a blowoff predictor, but rather a local extinction predictor—in particular, that each $\overline{Da}$ value is associated with a certain probability/event count of local extinction on the flame sheet.

3.3 Local Dynamics-Observations. We next consider more "local" characterization of these extinction processes. Mie scattering diagnostics were used to visualize a density-based boundary between product and reactants. Such an approach is less direct and spatio/temporally precise than, say CH PLIF or other reaction rate measurements, but provides some indicator of where breaks in the flame exist. As long as the flame is sufficiently removed from blowoff, the raw image has two regions of high and low particle density regions, which indicate the cold reactants and hot products, see Fig. 11. Holes in the flame are associated with regions of significantly lower gradient in seed density. The raw image is filtered and processed in order to determine an edge between regions of high and low particle density. A threshold level (40%) is selected by examining the probability density function of the intensity gradients of a stable flame image. Flame holes are defined as the broken area along the flame, where the gradient of particle densities falls below the threshold, see Fig. 11. An instantaneous isovorticity field is also plotted on the top of flame fronts. It shows that each flame hole is associated with a region of high vorticity. However, it should be pointed out that there is a time delay between flame extinction and when a hole is evident from the density. When local extinction occurs, the density will not change immediately but over a molecular and turbulent diffusive time scale.

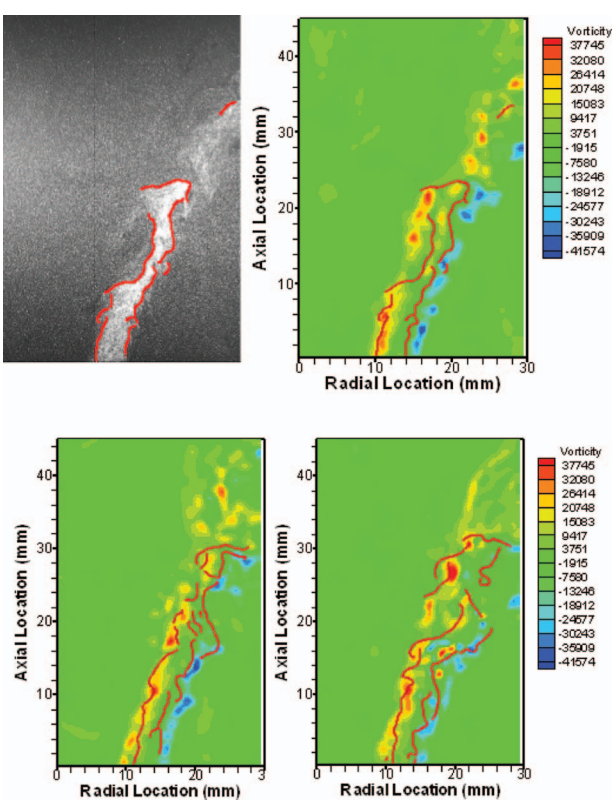

Fig. 11 Flame front in raw Mie scattering images (top left); Instantaneous isovorticity field and flame front (remaining images) of methane flame at equivalence ratio of 0.61

Figure 12 quantifies the spatial distribution of holes for a CH_4 flame. Out of 30 total images, the fraction of these that contained a hole (regardless of its size) in the inner flame branch was determined at each axial location from y to $y+\Delta y$, where $\Delta y=1$ mm. Interestingly, no holes are evident near the centerbody (2–5 mm downstream), even though this is the region of highest shear. There are two possibilities for this. One is that holes are initiated, but not yet evident in the Mie scattering intensity gradient, due to the finite time required for the temperature to smooth out. The second is that this is a region of product recirculation, which causes ignition of this region in the nearfield and which renders the flame difficult to locally extinguish. Similar situations occur in bluff body flames, where flame extinction usually initiates downstream and moves progressively closer to the bluff body as blowoff is approached (and opposite to nonpiloted jet flames which lift off the burner before blowoff) [28].

Moving downstream, the fraction of images with holes increases monotonically in a roughly linear manner. Note that the threshold used for flagging a hole does affect the location of the inferred hole. However, changing the threshold value shifts the linearly increasing region up or downstream downstream but not the basic shape of this plot. Also shown is data for a 50/50 H_2/CH_4 flame, which shows similar behavior.

3.4 Local Dynamics-Discussion. These data show that flame holes locations are not uniformly distributed along the flame—rather they increase monotonically with downstream distance. This behavior can be understood by noting that once a hole is formed, it is convected downstream at the local tangential velocity. The size of these holes will change with downstream distance, initially growing and then shrinking. However, if the time required for these holes to close is slow relative to the time in the viewing window ($\sim$20 mm/30 m/s=0.67 ms), then once formed, the hole will persist. Denote the probability of initiation of a hole per unit axial location and per unit time as p. If this probability is spatially uniform (i.e., equal likelihood of hole being initiated over entire viewing window), then the probability of a hole being present at a given axial location, x, per unit axial distance, $P(x)$, is simply

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$$P(x) = px/U_t \tag{7}$$

where U_t denotes the axial convection velocity of the hole. In other words, this equation shows that a spatially uniform probability of hole initiation implies a linearly increasing probability of a hole being present with axial location, x. Thus, the slope of the data shown in Figure 12 can be interpreted as the probability of hole initiation. The dashed line in the data is a fit to these data with a slope of $p=0.7$ hole initiation events/ms.

4 Concluding Remarks

We close this paper with some speculations regarding the processes controlling swirling flames near blowoff. Many of these comments are very similar to hypotheses raised in our bluff body work [28]. Lean blowoff occurs in at least two phases. The first is associated with spatiotemporally localized extinction events, leading to flame "holes." Such a phenomenon is largely independent of the recirculation zone dynamics. This study has shown that H_2/CH_4 mixtures reach this limit at similar global Damköhler numbers. The actual blowoff event is a more complex phenomenon that also involves interactions between the inner recirculation zone (vortex breakdown bubble), outer recirculation zone of the rapid expansion, and flame extinction/reignition phenomenon, which are not well understood. Some appreciation for the complexity of these "Stage 2" dynamics can be gained by studying the image sequences and discussion in the Ph.D. theses of Murugandam [48] and Zhang [49].

It appears that classical Damköhler number scalings based on average quantities are correlations for the first preblowoff stage where holes in the flame occur. That is, from a fundamental point of view, these global Damköhler number correlations do not describe the ultimate blowoff condition itself, but rather the condition at which flame extinction begins to occur. Figure 6 shows that these two limits are correlated: the lines representing these two limits are almost parallel with each other. Global Damköhler number scaling cannot describe, however, what is the critical $\mathfrak{I}$ value, $\mathfrak{I}_{\text{crit}}$, at which blowoff occurs; this critical value is likely influenced by the recirculating phenomenon that provides a "torch" for the near separation point region from which the flame then propagates out from. As such, the ability of $\overline{Da}$ correlations to describe/predict the actual blowoff condition is directly linked with the extent to which the ultimate blowoff event is correlated with these extinction events. Local extinction and global blowoff must be related, e.g., flame holes appear under near blowoff conditions and increase in frequency and duration as one gets closer to blowoff. Moreover, given the fact that average Damköhler number scalings can reasonably collapse blowoff data also indicates that they are related [44]. The problem, however, lies in the fact that the flame can withstand a certain amount of extinction, but not "too much," i.e., $0<\zeta<\zeta_{\text{crit}}$. It is clear from data that the flame can withstand some extinction, but that blowoff occurs well before the majority of the flame sheet is extinguished. However, determination of what constitutes this critical level of flamelet disruption and extinction is unclear at this point and is, we suggest, a key unsolved problem remaining to understand the blowoff phenomenology.

Nomenclature

D, d = diameter
Da = Damköhler number
S = swirl number
s = flame surface
S_L = laminar flame speed
U = flow speed
P = probability of presenting a hole
p = probability of initiation a hole
x, y = coordinate
α = thermal diffusivity
κ = strain rate
θ = swirl angle
σ = variance
τ = time scale
ζ = extinction fraction

Subscripts

chem = chemical
edge = flame edge
ESR = extinction strain rate
ext = extinction
flow = based on flow
PSR = perfectly stirred reactor
res = residence

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